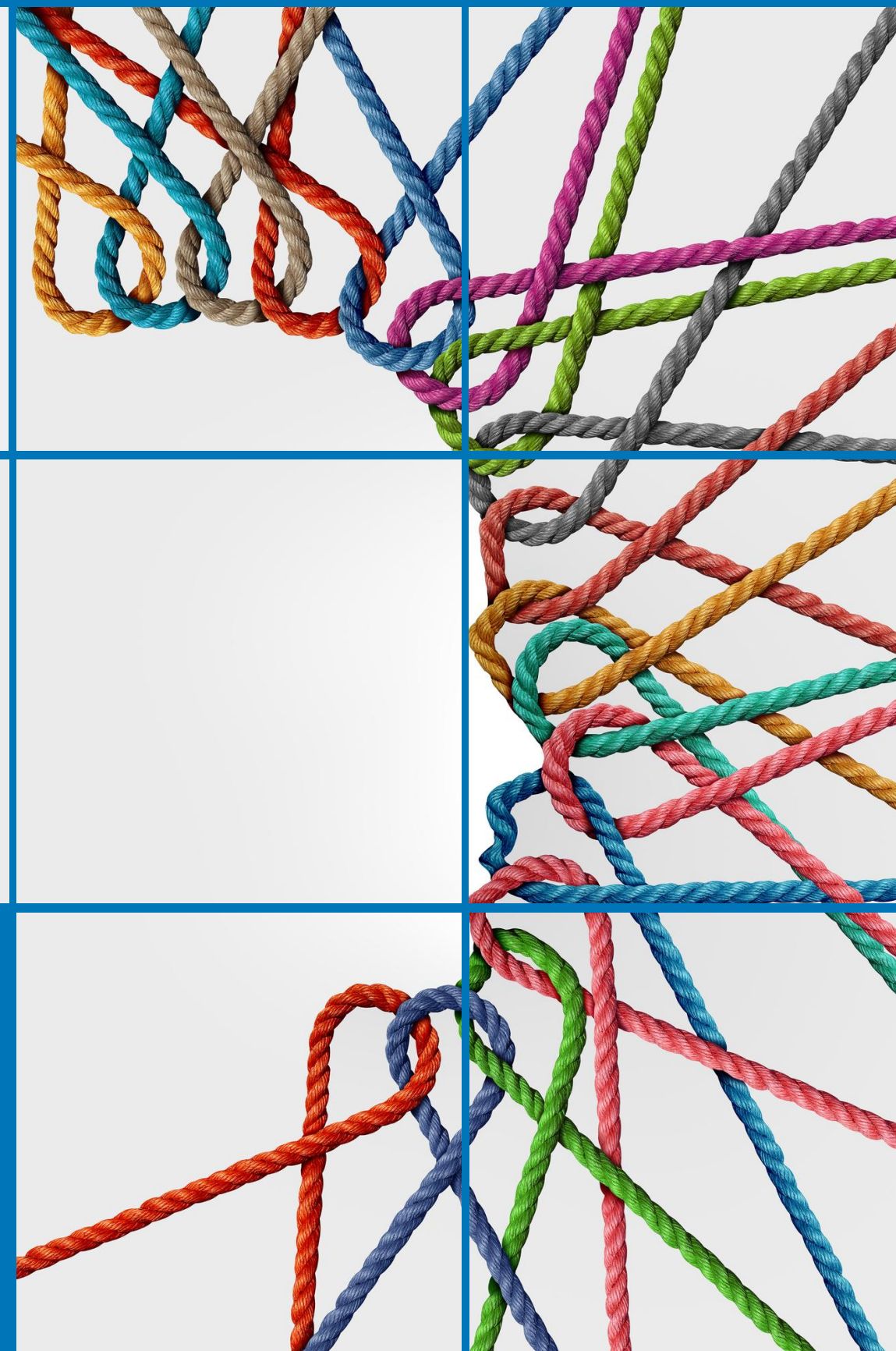




2026
March

FISERV UPI CONNECTOR RS Software Proposal

Technical, functional, deployment,
architecture and commercial walkthrough





From Vision to Victory: RS Software wins NASSCOM SME Inspire Awards

Tech Excellence in Financial Services | 2026

Reflects – the trust, commitment, & culture

Reinforces – deep expertise, discipline, future-ready solutions

Resolve – remain thoughtful, reliable, forward-looking partner to customers seeking proven expertise & innovation in payments

Kalpana Morparia (former CEO of JPMorgan India): power of agile & customer-centric value creation

S. Ramadorai (former CMD of TCS): excellence, adaptability, & character define lasting institutions

Why RS Software is the right partner for Fiserv's first build

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Leader introductions

02

Technical proposal rundown

Functional breadth, operating model, why solution is more than a thin connector

03

Accelerated deployment approach

Code readiness, accelerators, reusable assets, tactics that compress certification & launch timelines.

04

High-level architecture

Modular API-first architecture, integration points, active-active deployment posture.

05

Commercial walk-through

Build, operate, transfer, milestones, schedule



Introductions

Fiserv Leadership Panel



**Suresh
Chandrasekaran**

APAC Head Banking &
Fintech Solutions,
Singapore



Delvis Roy

Head of Technology,
Financial Solutions,
Pune



Devendra Trivedi

Business Leader,
Mumbai



**Sivakumar
Swaminathan**

Vice President - Head
of Technology,
Chennai



Rohini Joshi

Vice President &
Expert Engineer,
Singapore



Praveen Asare

Project Manager,
IT Infrastructure, Thane



**Vishal
Kadwadkar**

Sales Director, Mumbai



**Kesava
Guttikonda**

Vice President and Chief
Technology Officer, Singapore

RS Software's Leadership Panel



**Nitin Sudhakar
Thombare**

Director, Sales, India



Abhishek Gupta

Head, Global Pre
Sales & GTM Strategy



Abhishek Seal

Chief Architect, Products
& Innovation, NPCI UPI
Solution Leader



**Sujit Shankar
Banerjee**

Head, Global Product
Engineering, RS DigitalEdge™



**Deep Sankar
Saha Roy**

Head, Global Delivery &
Implementation Service



Prasenjit Maitra

Head Dev Lead



The right technical & delivery model for a production-grade, certification-ready UPI operating layer.

30–40%

faster time to market
through accelerators
and reusable assets

8–10 mo

tentative go-live
target after PO and
mobilization

99.99%

availability target
with active-active
deployment posture

Functional solution – comprehensive coverage

The proposal spans acquiring, issuing, merchant enablement, and production operations as a single UPI operating layer.



Payer PSP / customer initiation

- Device binding, registration, VPA lifecycle and account linking.
- P2P, P2M, collect approval, QR scan-and-pay and balance enquiry.
- Mandates and AutoPay lifecycle, disputes initiation, optional UPI Lite and eRupi.



Payee PSP / merchant acceptance

- Merchant and aggregator onboarding with verification controls.
- Static and dynamic QR generation, callbacks, search APIs and refunds.
- Disputes, merchant notifications, auto-refund logic and settlement metadata capture.



Issuing / bank processing

- CBS integration for debit, credit, reversal, refund and balance enquiry.
- UDIR, incomplete transaction handling, OTP and HSM-backed security.
- Mandate execution, credit line and clearing / settlement reconciliation where in scope.



Platform & ops

- Multi-tenancy, maker-checker, user access and scheduled jobs.
- Unified transaction state, dashboards, audit logs and reporting.
- Runbook-ready operations, notifications and support controls.

Why this matters for Fiserv

- *One reusable core instead of fragmented point solutions.*
- *Both bank and merchant ecosystems can be served from the same platform.*
- *Operational essentials such as repairs, reversals, disputes, reconciliation and auditability are built in from the start.*

	Pay	<i>P2P / P2M / P2PM</i>
	Collect	<i>request-to-pay</i>
	Mandate	<i>AutoPay lifecycle</i>

Delivers UPI Operating Layer

UPI delivery succeeds when transaction state, repair logic, and certification controls are treated as first-class design elements.

Common delivery risk if UPI is treated as a thin connector

Stateful transaction lifecycles
 Stateless forwarding cannot safely handle multi-step UPI outcomes.

Timeouts and ambiguous outcomes
 Pending outcomes create customer, settlement and dispute exposure.

NPCI semantics and interface rigor
 Frequent scheme updates and response-code behavior create rework and instability.

Operational traceability and audit
 Operations teams cannot manage disputes or audits without durable keys and lifecycle history.

How the RS proposal neutralizes that risk

Stateful transaction lifecycles
 A central UPI core manages state machines, idempotency, routing, and terminal outcomes across Pay, Collect and Mandate.

Timeouts and ambiguous outcomes
 Automated status enquiry, repair queues, reversal flows, and reconciliation-led correction are designed into the platform.

NPCI semantics and interface rigor
 Reusable adapters isolate message formats, signing, encryption, key handling, and participant synchronization.

Operational traceability and audit
 Unified transaction history, reconciliation keys, dashboards, and dispute linkage support production operations from day one.



Faster Deployment:

Time compression comes from reusable assets, early integration, and parallel certification discipline rather than from cutting scope.

What is already accelerated before bespoke build begins -

- API-first microservices foundation with reusable real-time payments components.
- NPCI simulators, stubs and test harnesses to front-load integration and defect discovery.
- Adapter patterns for CBS / issuer host, notifications, HSM and merchant systems.
- Configurable workflows, rules, limits and maker-checker controls to reduce custom code volume.
- Observability, correlation IDs, security scaffolding and operational controls already patterned.
- Reusable regression assets and certification-oriented delivery factory to reduce rework.

Deployment levers that shorten the critical path

Baseline platform and core integrations are established before full flow completion.

Benchmarking and certification activities run in parallel once the platform is stable

Mock components protect the plan from HSM or downstream system readiness delays

Phased UAT drops create earlier business feedback and cleaner acceptance gates.

Message for the call: speed comes from readiness and reuse, not from compromising production hardening.

M6–9

parallel certification window

30–40%

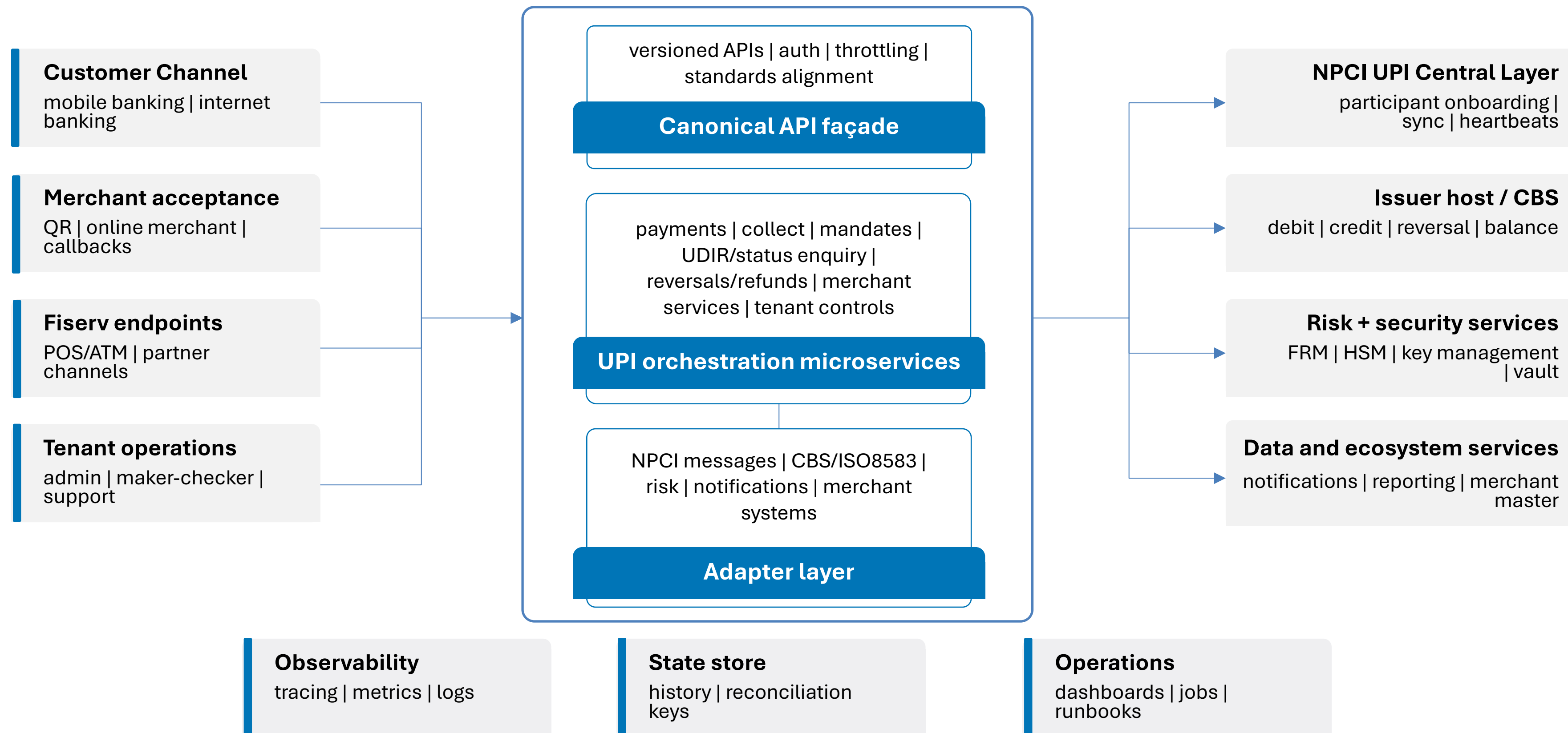
faster time to market

3–4 wks

mobilization after PO

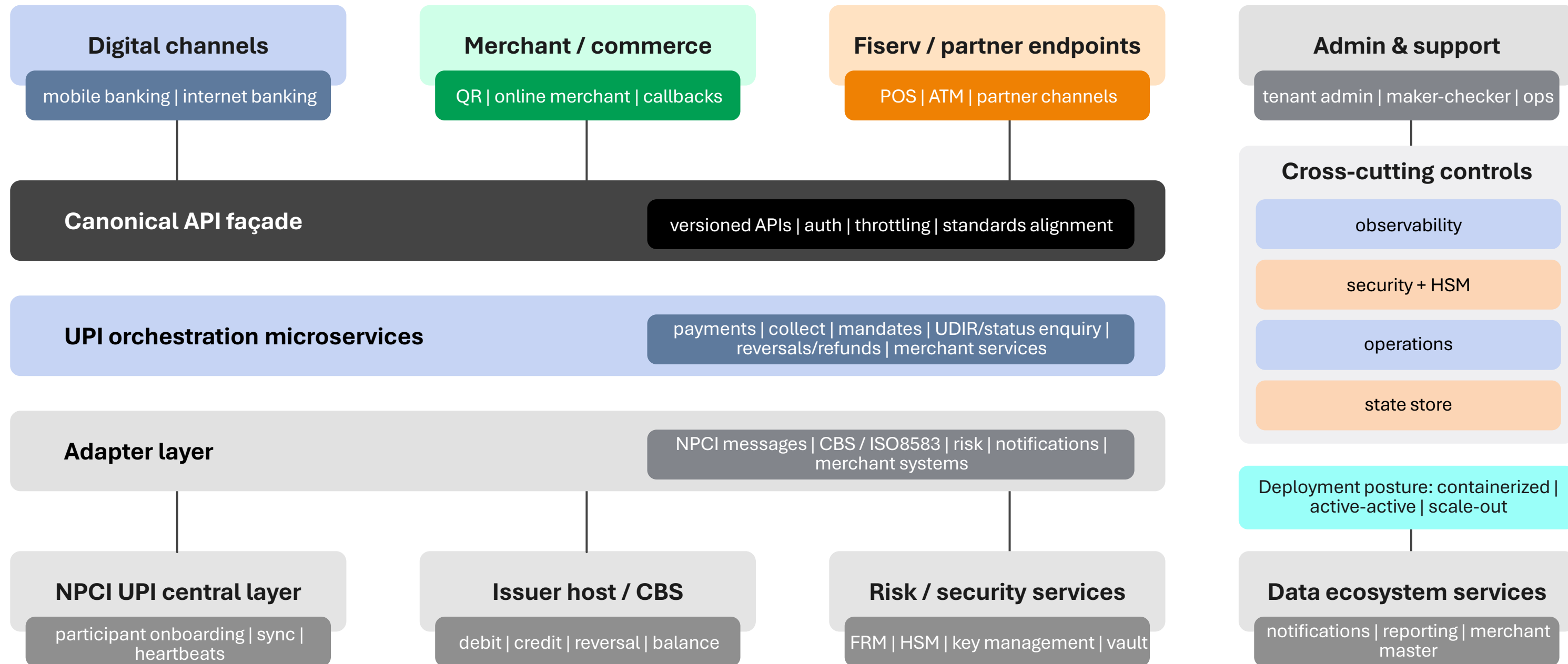
Solution Overview

A single UPI core isolates channel complexity, external interfaces, and production operations while remaining deployment-flexible.



High-level architecture layout

A modular, API-first, active-active design isolates channel complexity, transaction orchestration, and enterprise integrations.



99.99%

availability target

Active-active multi-site design with redundancy through every layer and rapid failover.

Low ack latency

performance target

Latency-sensitive APIs stay synchronous while non-critical activities move to asynchronous processing.

Linear scale-out

scalability target

Containerized services scale horizontally to support growth and traffic spikes.

Financial truth

data resiliency

The proposal targets resilient cross-site financial state without relying on synchronous replication bottlenecks.

Production readiness and NFR posture

Indicative targets are already expressed in the proposal; discovery will validate sizing and finalize thresholds.

Observability and control plane

- Correlation IDs and distributed tracing across channel, façade, services, adapters, and external systems
- Structured logs, health checks, SLA dashboards, and synthetic monitoring
- Standard error taxonomy, retries, throttling, and circuit breakers for downstream resilience

Security and compliance posture

- Correlation IDs and d• HSM-backed cryptography, key rotation, certificate lifecycle controls, and vault-managed secrets
- Least-privilege access, RBAC/ABAC, strong privileged workflows, immutable audit logs
- Maker-checker governance and policy-driven retention for sensitive operational changes

Configurable business rules

- Tenant-specific limits, routing, risk overlays, merchant properties, and exception handling
- Backward-compatible API versioning and contract testing to reduce impact on channels and commerce platforms
- 24x7 monitoring, DR drills, release management, and runbook-backed support readiness

Delivery roadmap: phased build with early readiness and parallel certification

The proposal targets an 8–10 month go-live by running readiness activities in parallel once the baseline platform is stable.

Phase 2 M3–7	Phase 4 M8–10	Phase 0 M0–1	Phase 1 M1–3	Phase 3 M6–9
UPI flows and bespoke enhancements	Go-live and hypercare transition	Discovery and solution definition	Platform baseline and core integrations	NPCI certification and readiness
• Pay, Collect and Mandate • Merchant callbacks, auto-refunds, disputes • E2E, performance and resilience testing	• Production cutover and pilot ramp • KPI monitoring and tuning • L2/L3 transition and release cadence	• Target operating model • Architecture, security and data flows • Certification plan and test strategy	• Environments and baseline services • CBS / HSM / notifications / merchant integrations • Tenant controls and operational foundation	• Certification execution and defect triage • DR, SOPs, training and cutover planning • Parallel run to protect the critical path

Schedule protection mechanism: benchmarking and certification are intentionally overlapped after the platform baseline, reducing late-stage rework and shortening the path to launch.

Execution model: governance, dependencies, risk control

The proposal is credible because it identifies what Fiserv must provide, how decisions are governed, and where the critical risks sit.

Governance cadence

- Steering committee (bi-weekly): executive oversight, scope, timeline, escalations
- Program management (weekly): plan tracking, RAID, dependencies, financials
- Engineering squads (2–3 times/week): design reviews, sprint planning, integration, defect triage
- Certification war-room (as needed): daily checkpoints during certification cycles

Operating model options:

- Build-Operate-Transfer
- Co-development
- Managed services for L3 and scheme updates

Dependencies on Fiserv

- Provide DEV / UAT / PRE-PROD / PROD environments and compatible CI/CD tooling
- Finalize hosting model, network zones, container platform, observability stack, and image repository
- Supply interface specifications and non-production endpoints for CBS, merchants, notifications, and digital channels
- Own VAPT and UAT execution while RS supports defect fixes
- Ensure HSM readiness, NPCI connectivity prerequisites, certificates, whitelisting, and operational procedures

Top risks and mitigations

1. Certification timelines and re-test cycles
Mitigation: early collaboration and dedicated checkpoints
2. Downstream CBS latency or instability
Mitigation: asynchronous patterns, throttles, circuit breakers, tuning, and joint performance testing
3. HSM / key-management delays
Mitigation: early design workshop, mock HSM, clear readiness checklist
4. Merchant callback failures and disputes
Mitigation: configurable callbacks, retries, auto-refund controls, monitoring, and reconciliation

Commercial walk-through: proposal structure and pricing summary

\$1.525M

build fee before
concession

\$300K

operate fee for year-1,
24x7x365, L2/L3 support

\$50K

knowledge transition,
train-the-trainer

\$977K

build fee with concession if PO
by Mar 26, 2026

Commercial structure

- Build: milestone-based design, development, certification and go-live over approximately ten months.
- Operate: 12 months of production support for stabilization and steady-state operations.
- Transfer: Train the trainer model. Up to 8 weeks of knowledge transition covering architecture, monitoring, incidents and troubleshooting.

Payment milestones – Build

Milestone	Trigger	What it covers	Fee
Signing amount	PO issuance	Mobilization and program start	20% of Build Fee
Phase 0	Discovery completion	Requirements and solution definition	15% of Build Fee
Phase 1	Core platform completion	Issuer, acquirer/payee, payer PSP and core integrations	20% of Build Fee
Phase 2	Flow delivery completion	Autopay, disputes, credit line, contextual payment, UPI Circle, Hello UPI, 123Pay, FIR/FOR, clearing & settlement	25% of Build Fee
Phase 3	Certification completion	NPCI certifications and readiness gates	10% of Build Fee
Phase 4	Production roll-out completion	Production deployment and ready for live/pilot transaction	10% of Build Fee

Payment milestones – Operate Support and Transfer

Milestone	Trigger	What it covers	Fee
Operate	Start of support year	24x7x365 L2/L3 production support for 12 months	100% of Operate Fee
Transfer	Start of KT period	Knowledge transition and train-the-trainer	50% of Transfer Fee
Transfer	End of KT period	Knowledge transition and train-the-trainer	50% of Transfer Fee

- Net 30 invoicing
- Fiserv to bear travel costs
- Change management applies to Fiserv-caused delays,
- Price adjustment applies beyond the stated long-stop dates

Fiserv take-away



- The proposal is a production-grade UPI core
- Functional coverage spans acquiring, issuing, merchant enablement, and operations.
- Delivery speed is supported by reusable assets, readiness discipline and certification planning.
- Commercials map cleanly to phased value delivery.
- Being built by the team that built UPI – Central Market Infrastructure



**THANK
YOU**